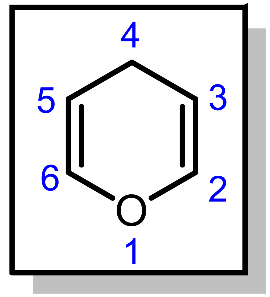
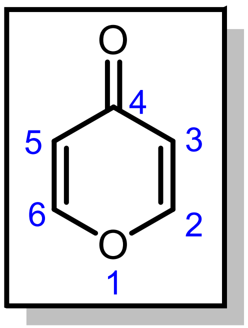
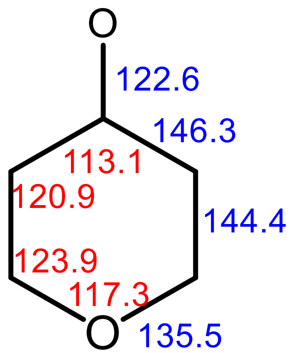
5.1.7. 4*H*-Pyran und Derivate
5.1.7.1. Stammsystem

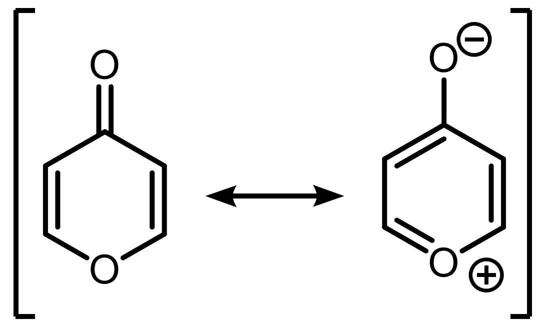


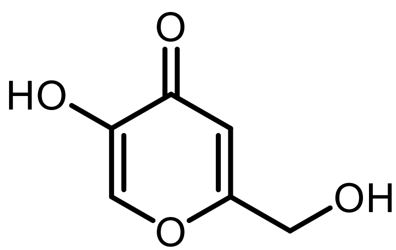
5.1.7.2. 4*H*-Pyran-4-on



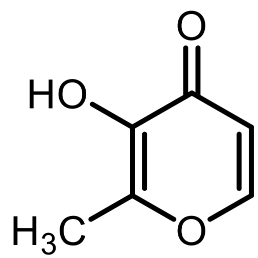
Bindungslängen [pm]
Bindungswinkel [°]



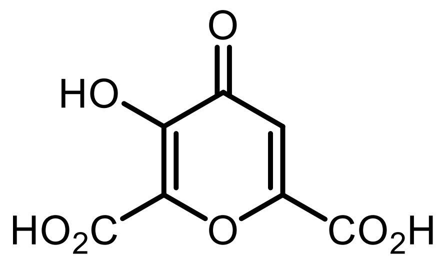




Kojisäure
(aus Zucker durch Bakterien)

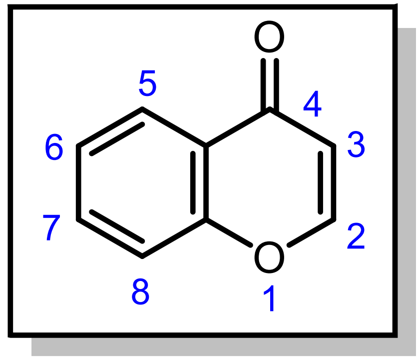


Maltol
(aus Lärchenrinde)

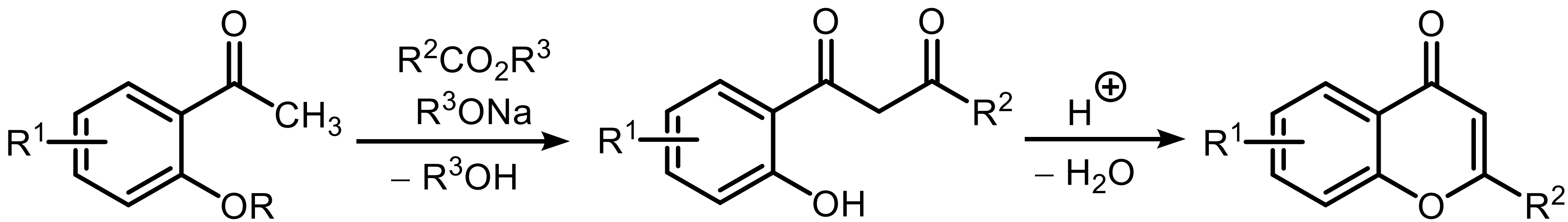


Mekonsäure
(häufig Salze der Opiate)

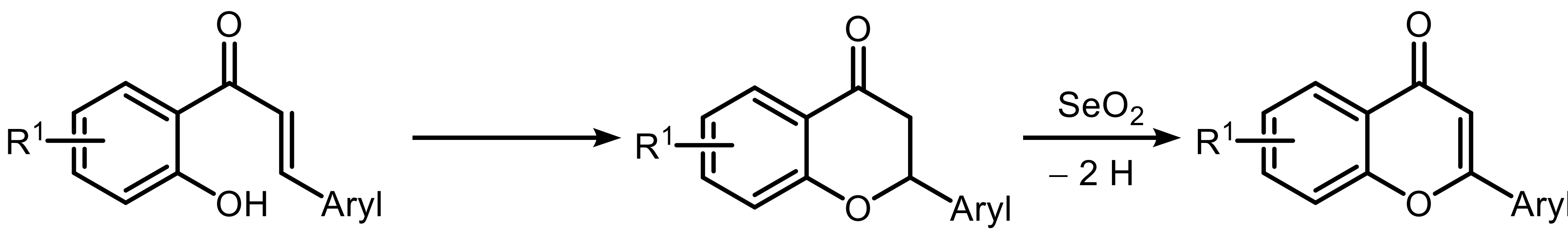
5.1.7.3. 4*H*-Chromen-4-on (Chromon)



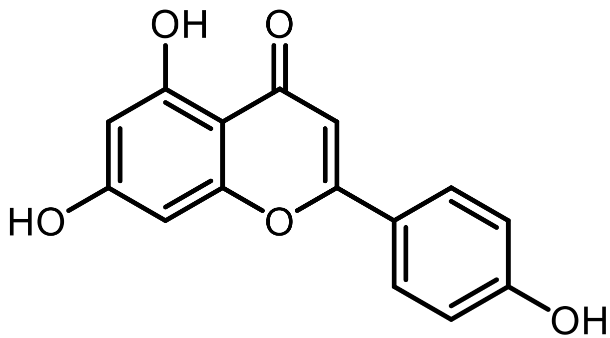
Chromon-Synthesen (ausgehend von *o*-Hydroxyacetophenonen)



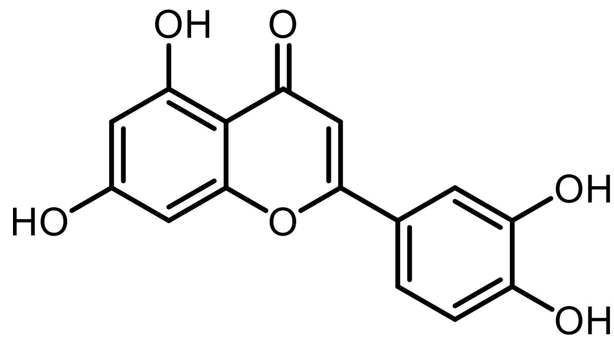
R = H, $SiMe_3$



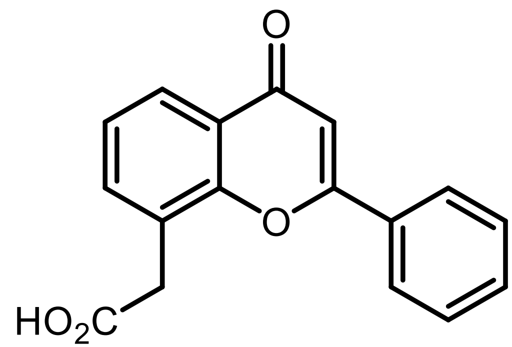
Chalkon
(durch Aldolkondensation)



Apigenin

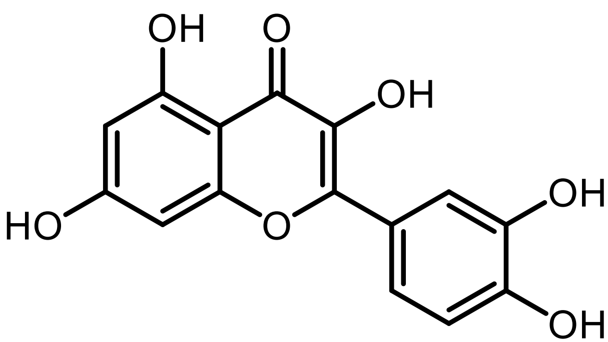


Luteolin

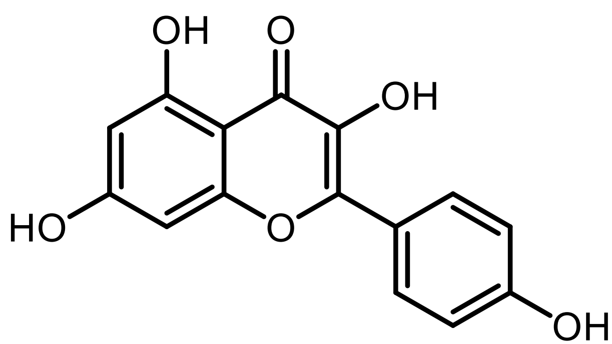


Flavon-8-essigsäure
(cytotoxisch, antitumor)

Flavonole

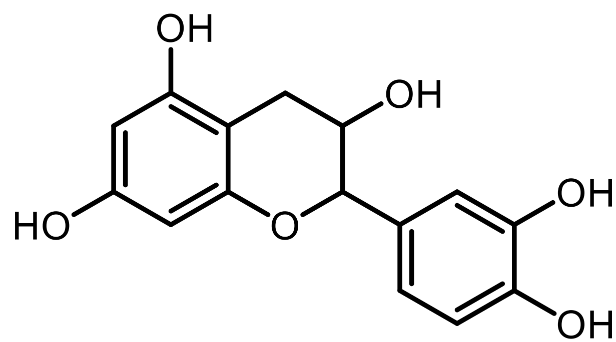


Quercetin



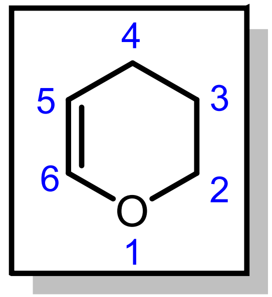
Kämpferol

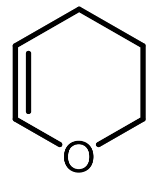
Flavanole



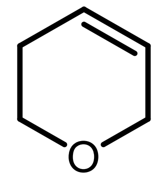
Catechin

5.1.8. Di- und Tetrahydropyrane und Derivate
5.1.8.1. 3,4-Dihydro-2*H*-pyran





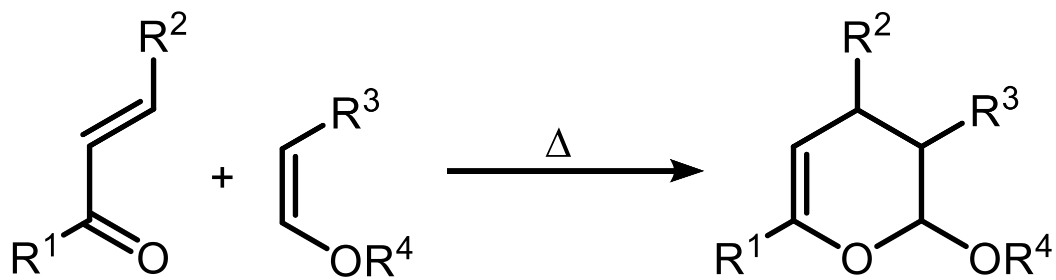
3,4-Dihydro-2*H*-pyran

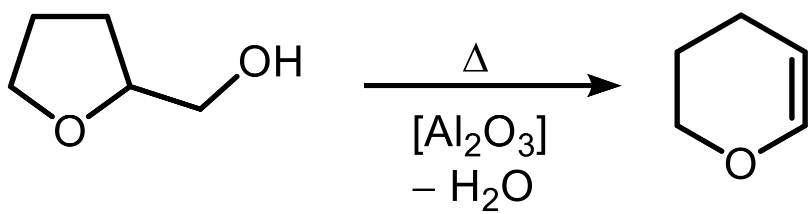


5,6-Dihydro-2*H*-pyran

Synthese

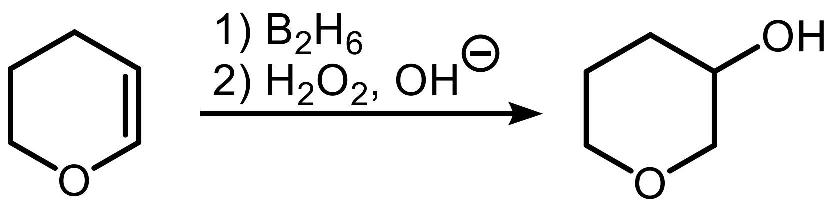
91



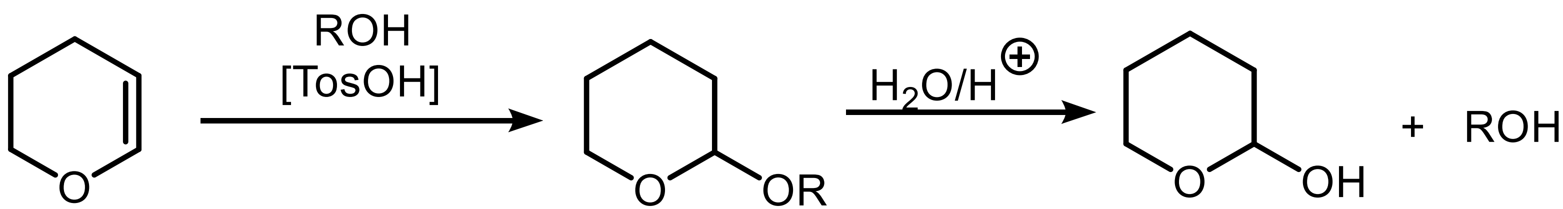


Reaktionen

92

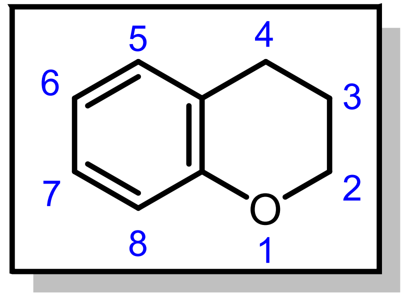


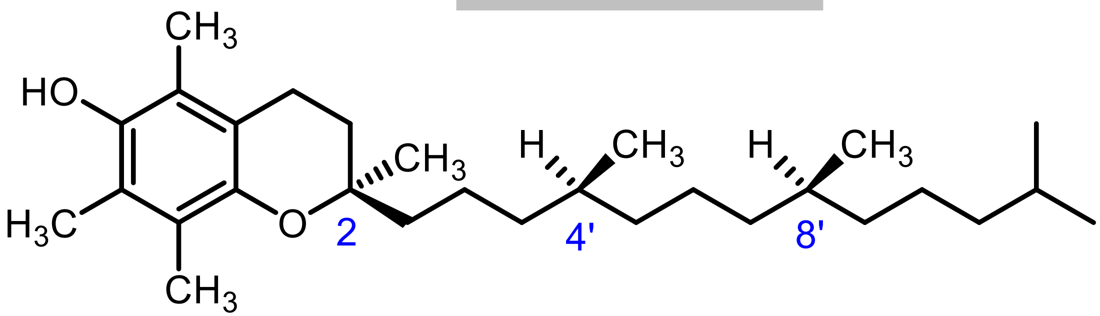
93



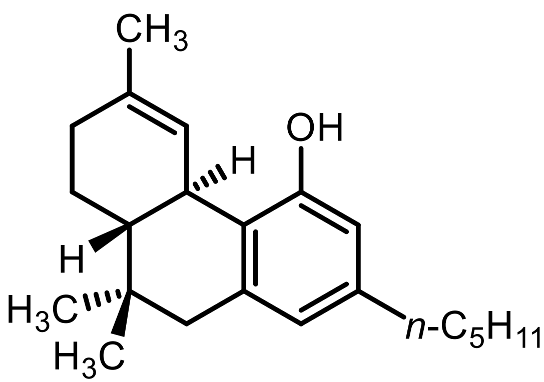
Bedeutung von DHP (3,4-Dihydro-2H-pyran):
OH-Schutzgruppe THP (Tetrahydropyran); allerdings Diastereomerenbildung bei chiralen Alkoholen

5.1.8.2. Chroman (Dihydrobenzopyran)

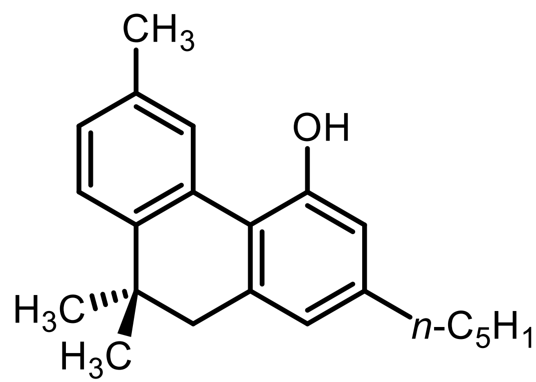




α-Tocopherol (Vitamin E)



Δ^9-*trans*-Tetrahydrocannabinol (THC)

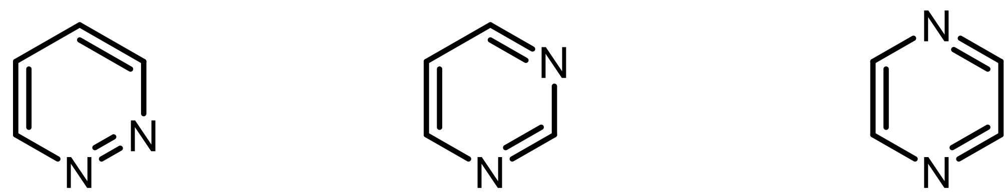


Cannabinol (CBN)

5.2. Sechsring-Heterocyclen mit zwei Heteroatomen

5.2.1. Diazine

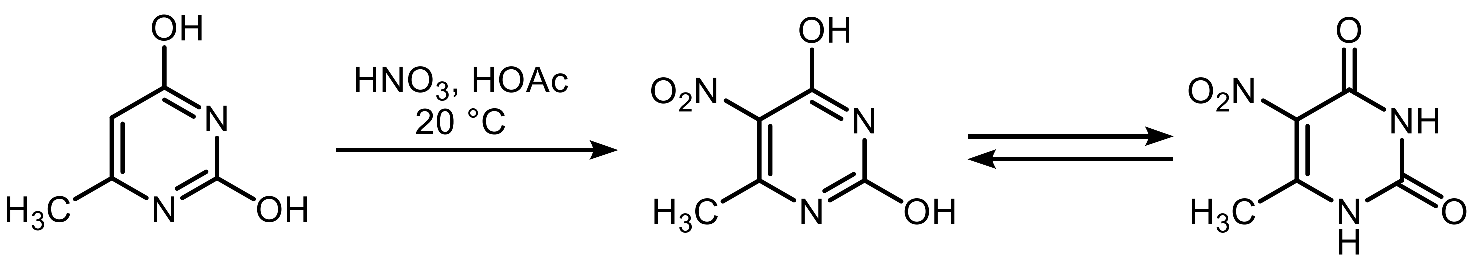
89



Pyridazin	Pyrimidin	Pyrazin
pK_S 2.3	pK_S 1.3	pK_S 0.5

5.2.1.1. Pyrimidin

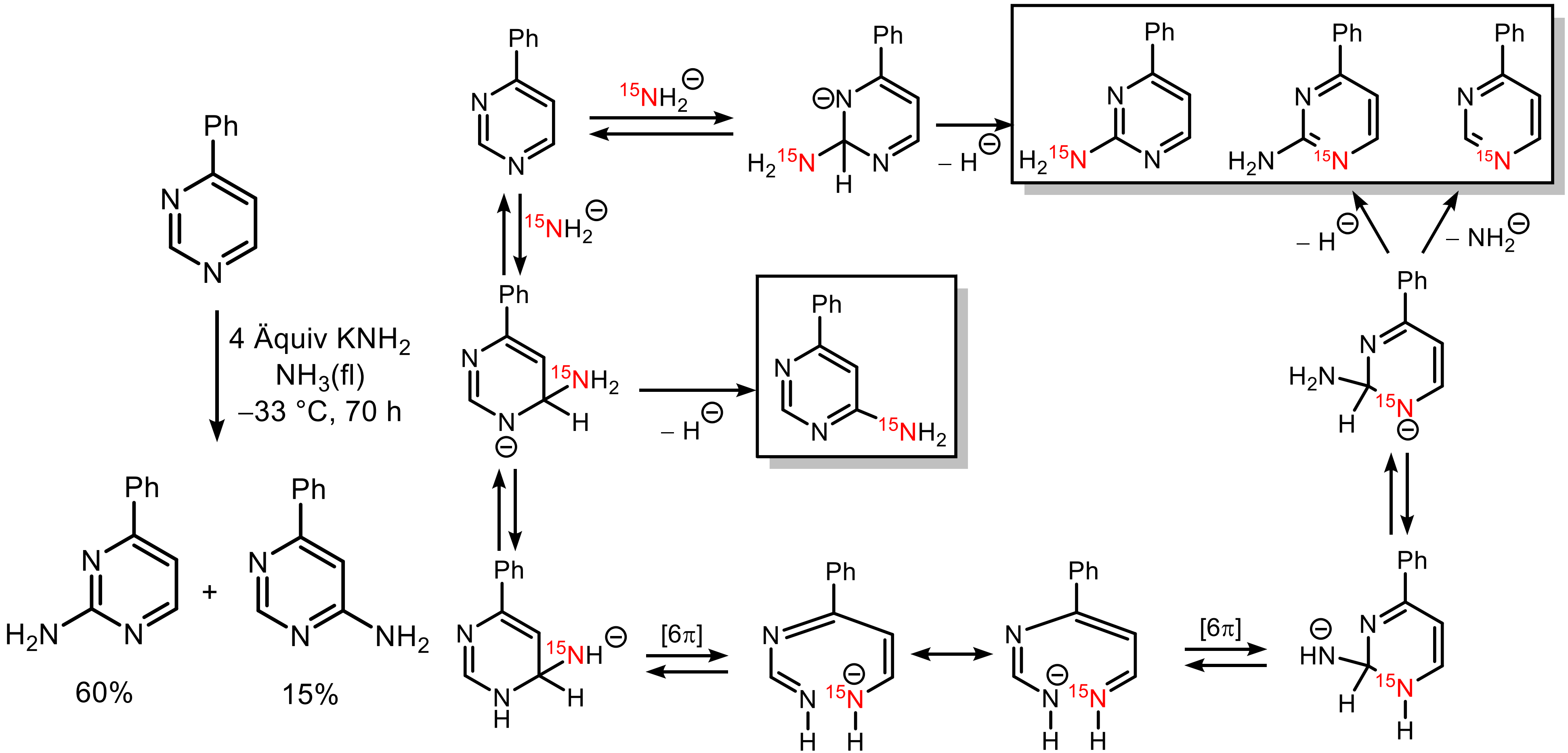
S_EAr: geringe Reaktivität (ähnlich 1,3-Dinitrobenzol, 3-Nitropyridin) steigerbar durch elektronenliefernde Substituenten. Nucleophile Substitutionen besonders bei 2- oder 4-Halopyrimidinen.



Nitrierung
Nitrosierung
Aminomethylierung
Azokupplung

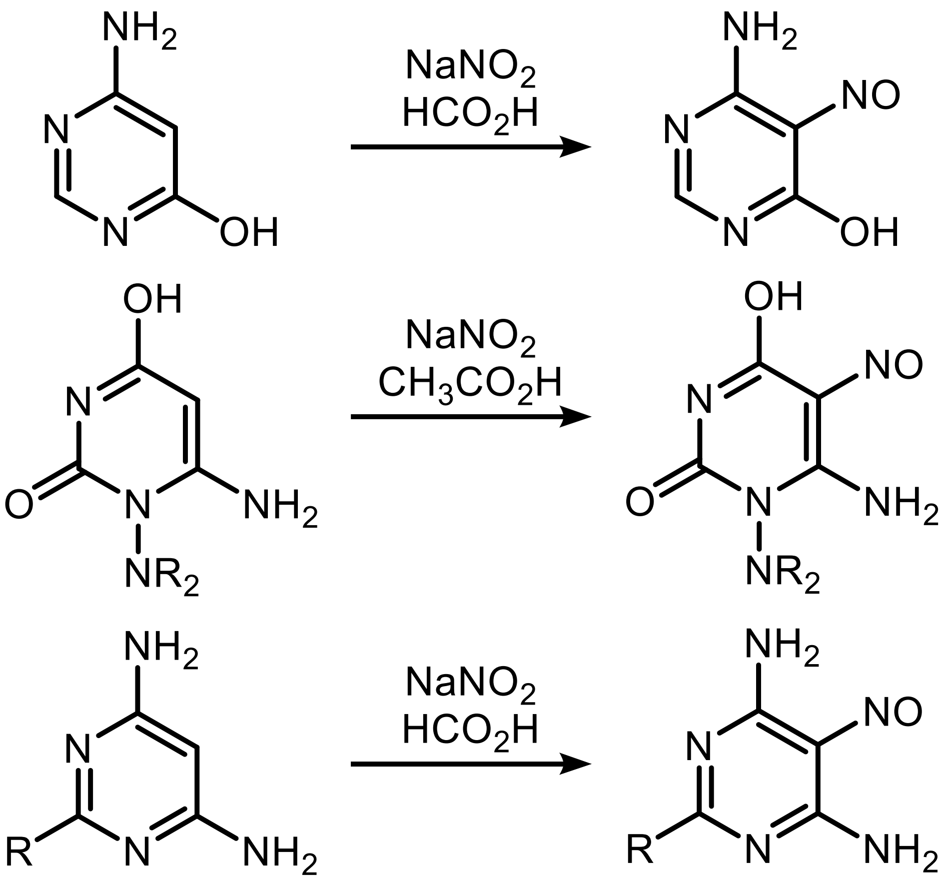
Reaktivität wie Benzol
S_NAr bevorzugt an C4 und C6 (z.B. Organometallverbindungen)

Tschitschibabin-Reaktion von Pyrimidin

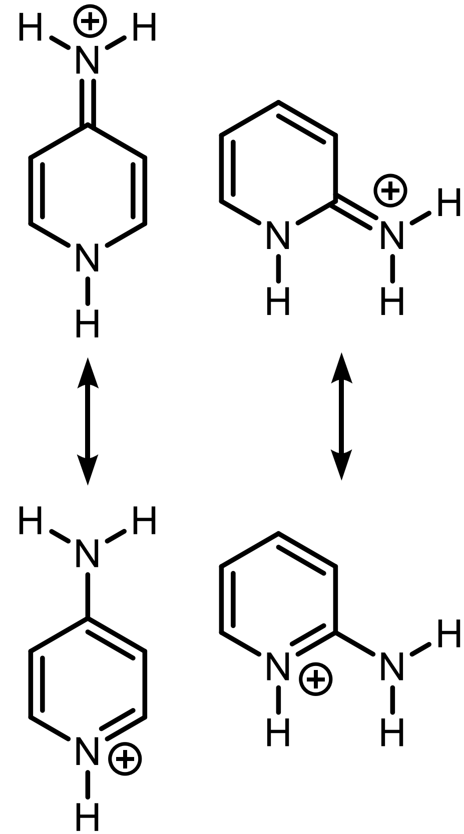


van der Plas, *J. Org. Chem.* **1979**, 4678.

5.2.1.1. Pyrimidin
Ring-Nitrosierung in Konkurrenz zur Diazotierung



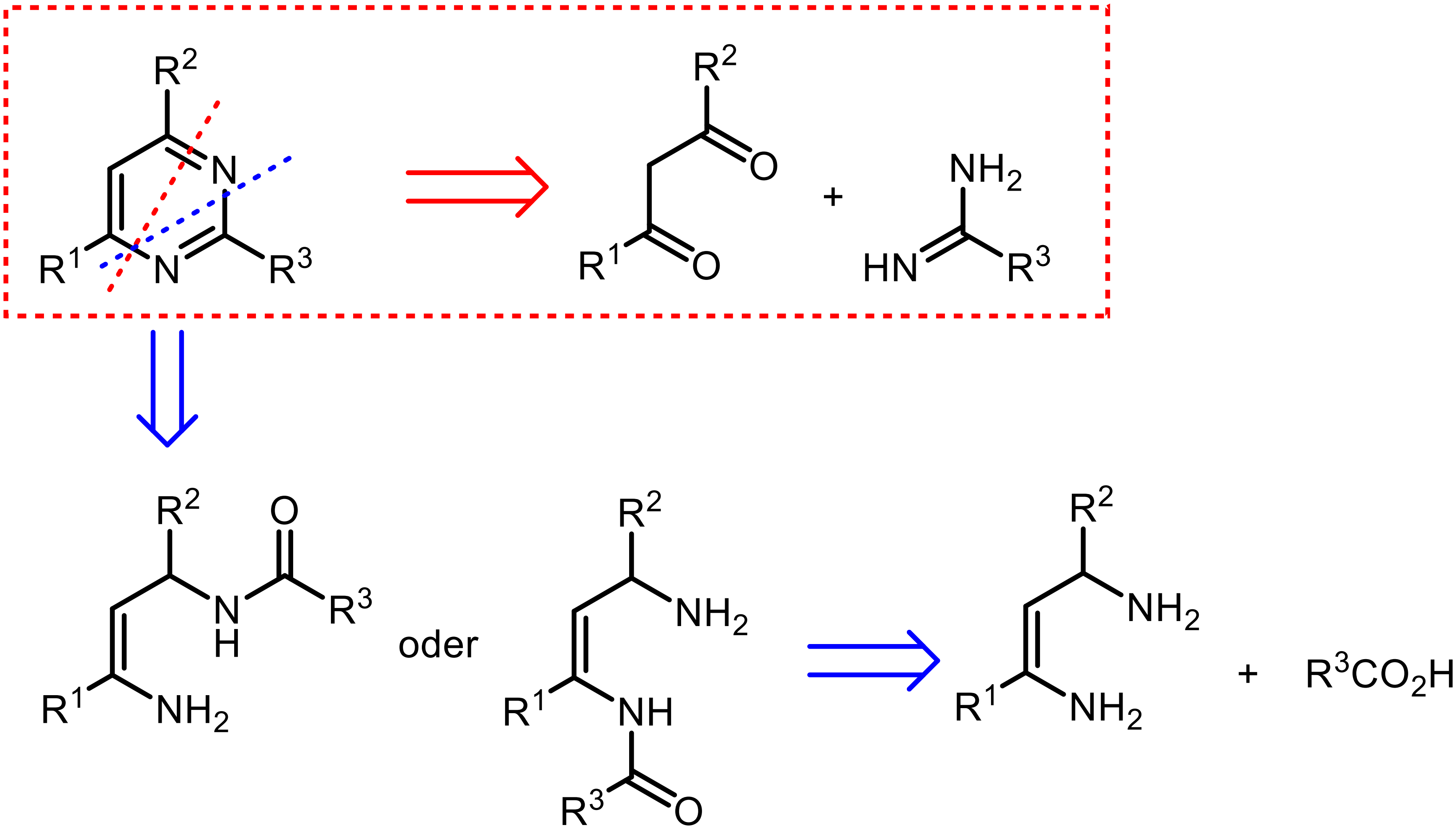
Diazotierung ist immer dann gegenüber der Ring-Nitrosierung erschwert, wenn sich die Aminogruppe in 2- oder 4-Position zu einem Ring-Stickstoff befindet.



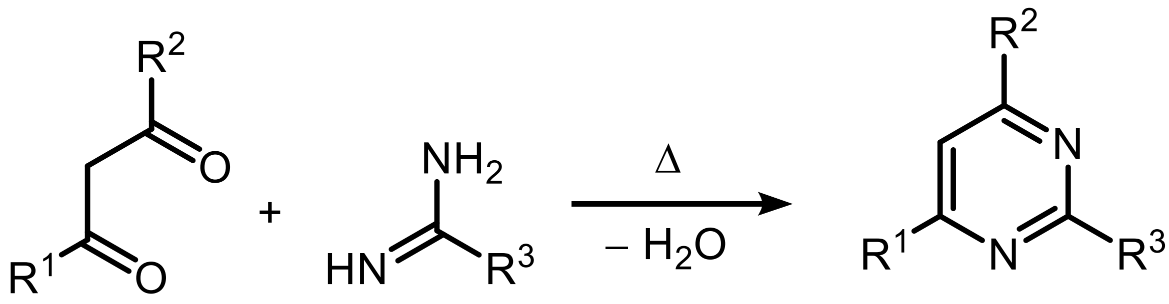
Butler, *Chem. Rev.* **1975**, 241.

5.2.1.1. Pyrimidin

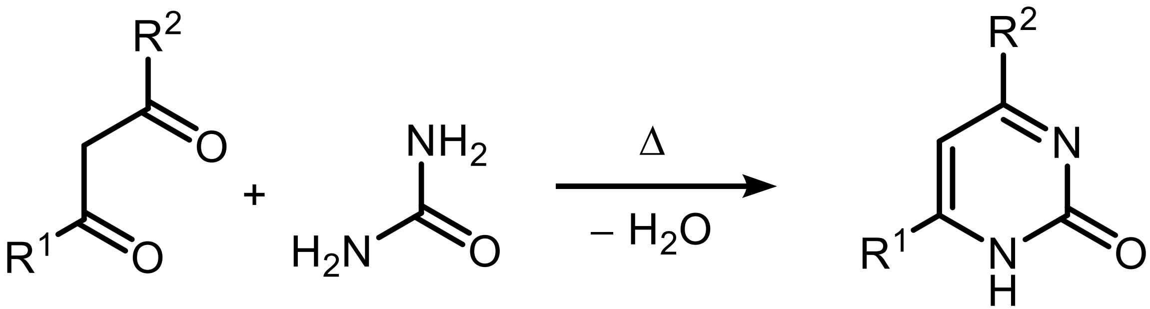
Retrosynthese



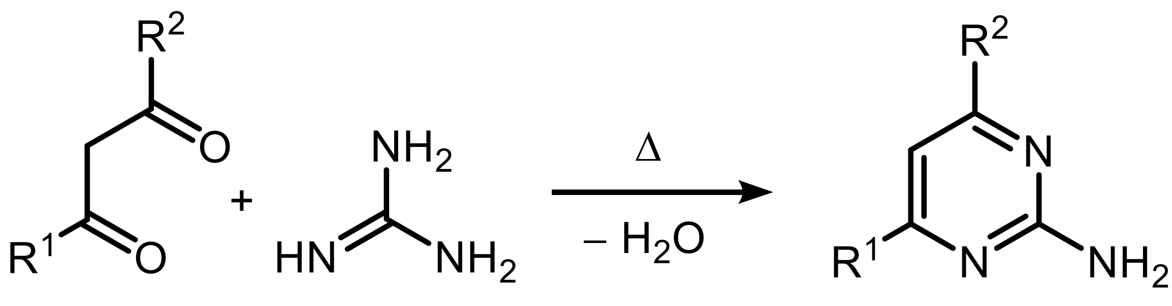
1) Pinner-Synthese



Pyrimidine



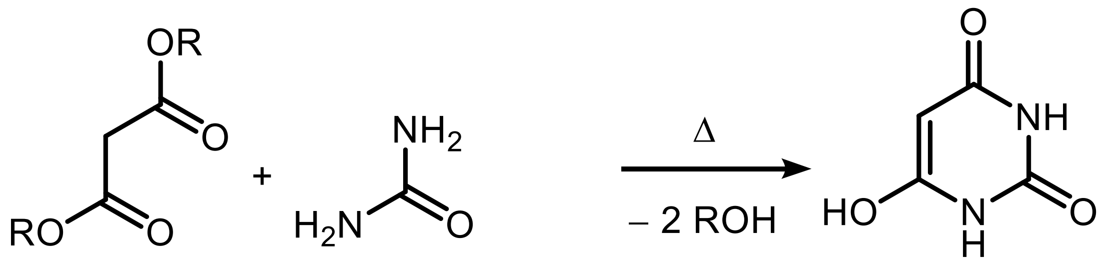
2-Pyrimidone



2-Aminopyrimidine

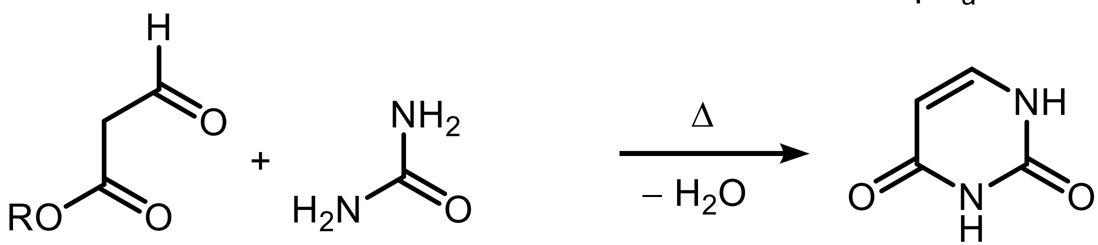
5.2.1.1. Pyrimidin

Pinner-Synthese

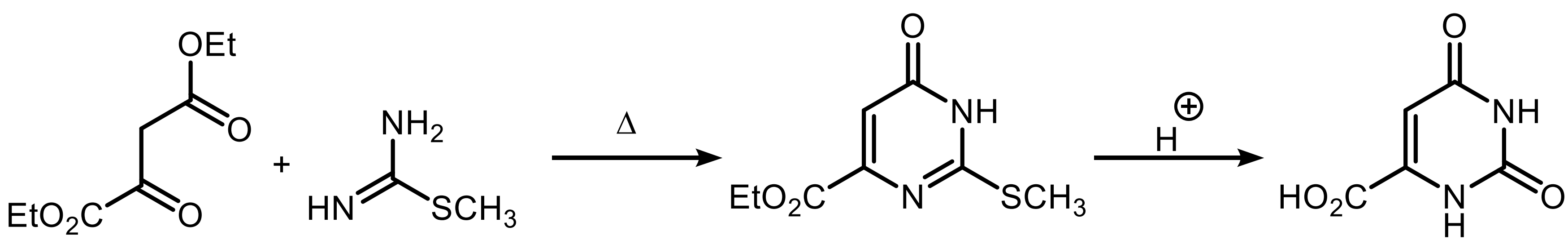


Barbitursäure

pK_a = 4.0



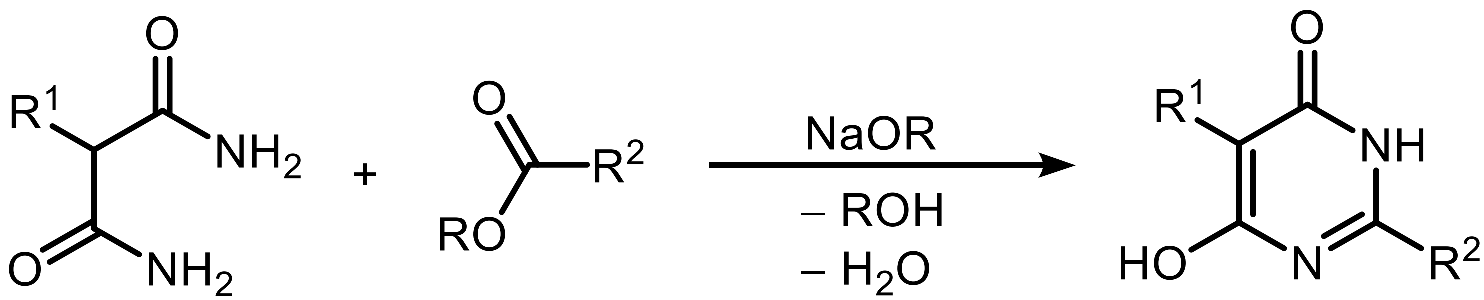
Uracil



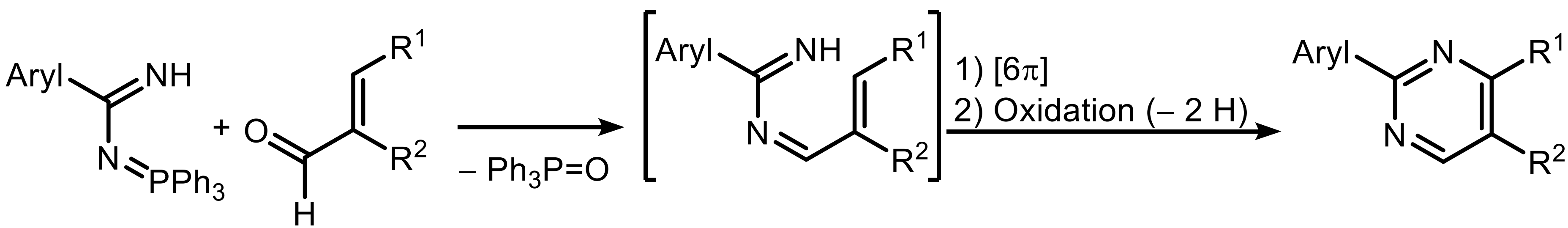
Orotsäure

(biochemische Zwischenstufe aller Pyrimidinderivate)

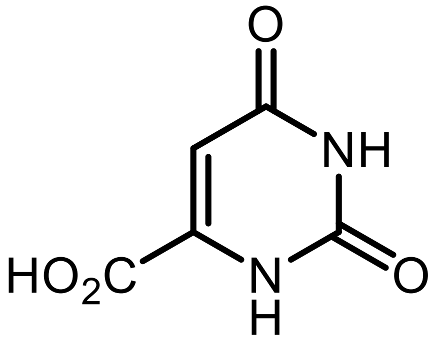
2) Remfry-Hull-Synthese

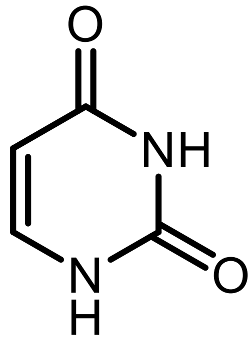


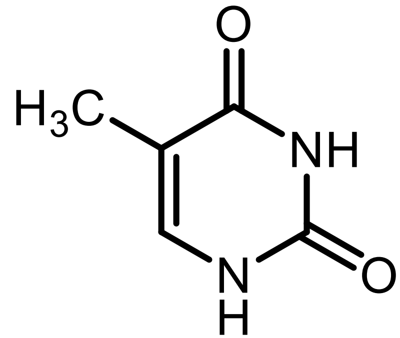
3) Über Aza-Wittig Reaktion

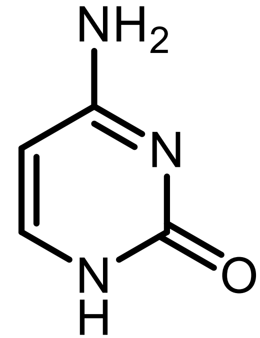


Vorkommen und Bedeutung







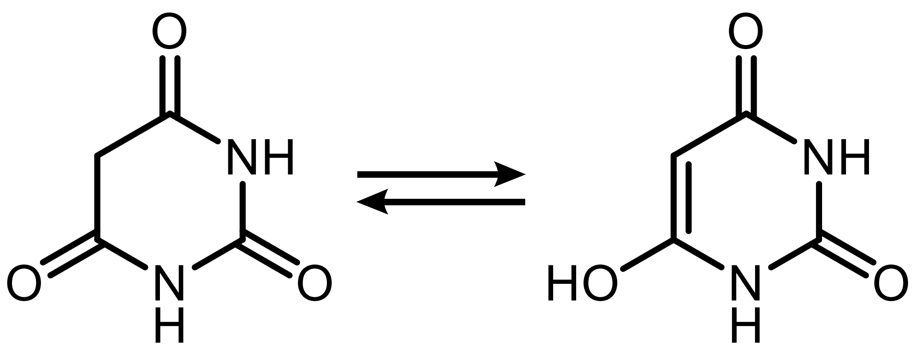


Orotsäure

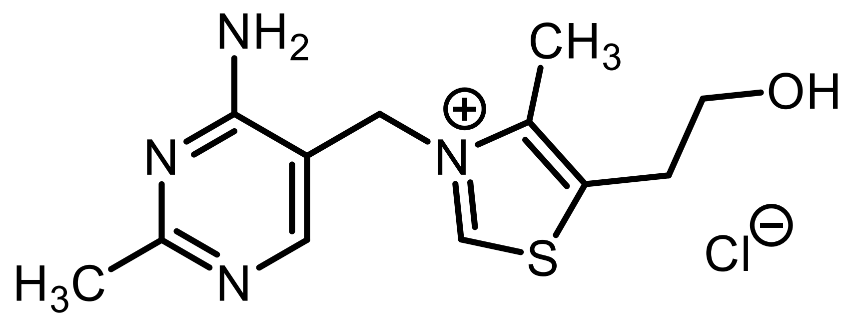
(Biosynthesebaustein)

Uracil Thymin Cytosin

(Pyrimidinbasen in Nucleinsäuren)



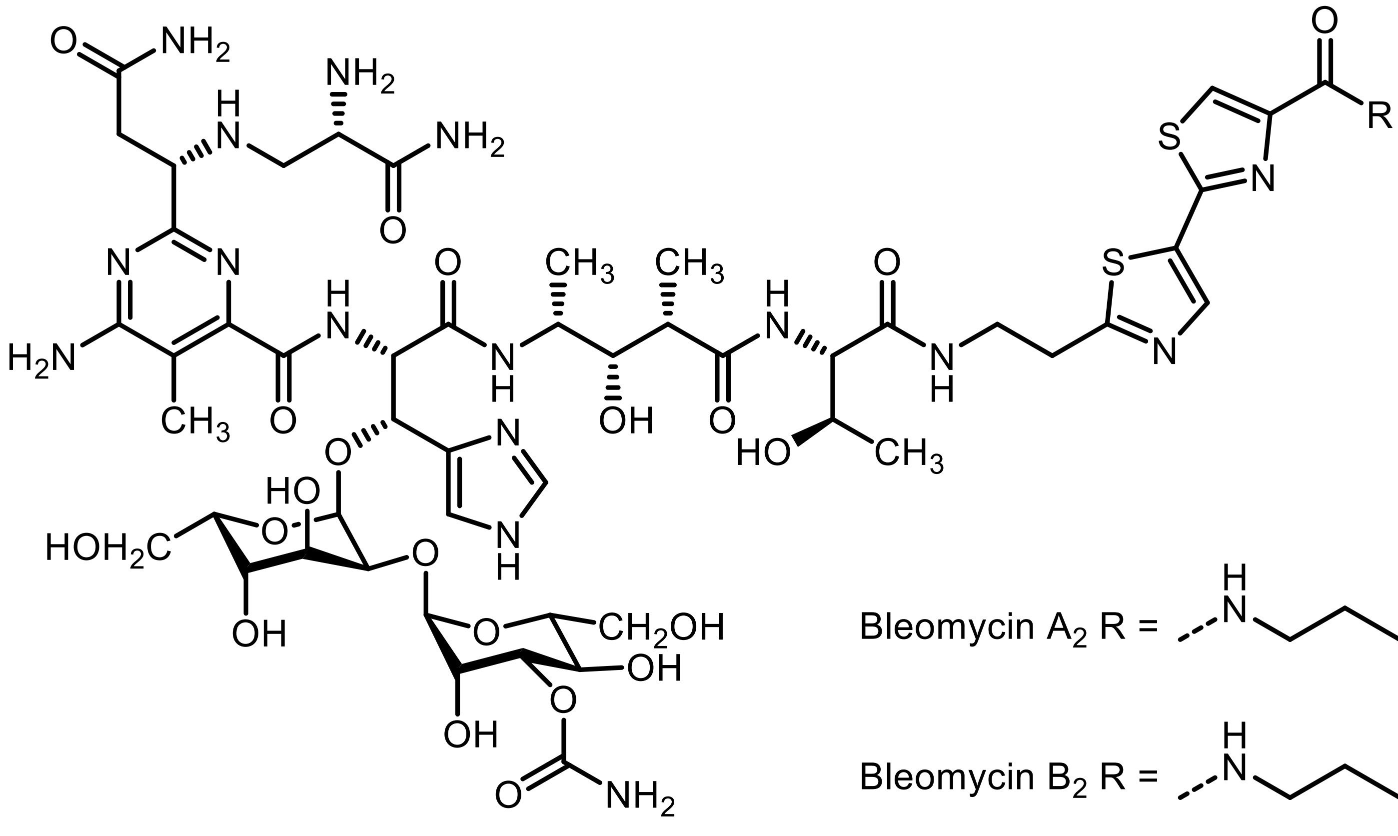
Barbitursäure

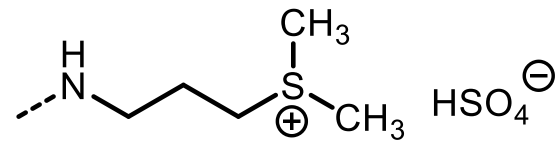


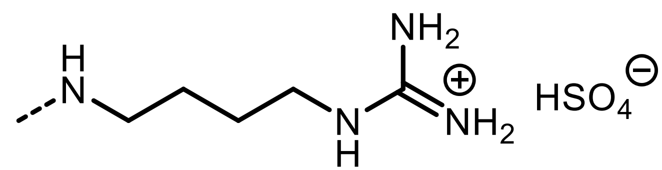
Aneurin

(Thiamin, Vitamin B_1)

Naturstoffe

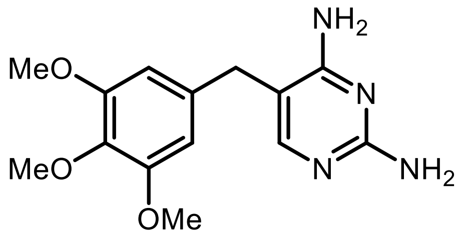


Bleomycin A_2 R = 

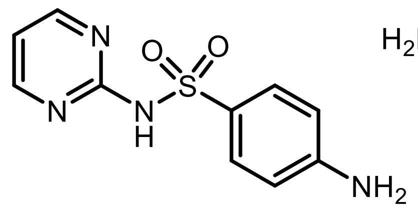
Bleomycin B_2 R = 

Tumorantibiotika
(von Streptomyces verticillus gebildet)

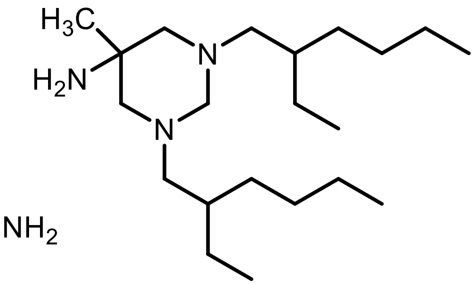
5.2.1.1. Pyrimidin
Wirkstoffe



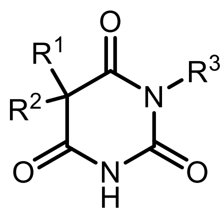
Trimethoprin
(Bakteriostatikum)



Sulfadiazin
(Antiparasitikum)



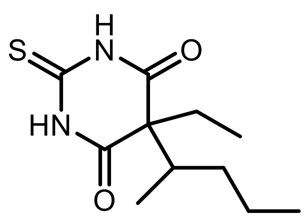
Hexetidin
(Antiseptikum im Rachenraum)



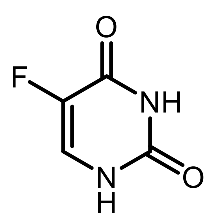
Barbiturate

$R^1 = R^2 = C_2H_5$, $R^3 = H$	**Veronal**, Barbital
$R^1 = C_2H_5$, $R^2 = Ph$, $R^3 = H$	**Phenobarbital**
$R^1 = CH_3$, R^2 = 1-Cyclohexenyl, $R^3 = CH_3$	**Hexobarbital**
$R^1 = C_2H_5$, $R^2 = Ph$, $R^3 = CH_3$	**Methylphenobarbital**

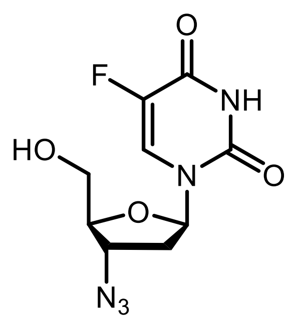
Hypnotica,
Sedativa,
Anxolytica,
Narcotika



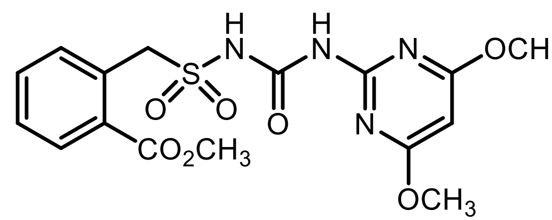
Thiopental
(Kurznarcotikum)



5-Fluoruracil
(Cytostatikum)



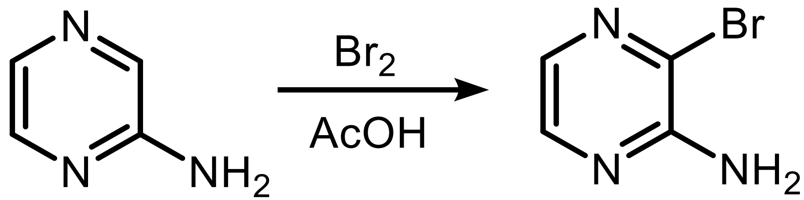
Zidovudin
(antivirales Chemotherapeuticum bei HIV-Behandlung)



Bensulfuronmethyl
(Herbizid)

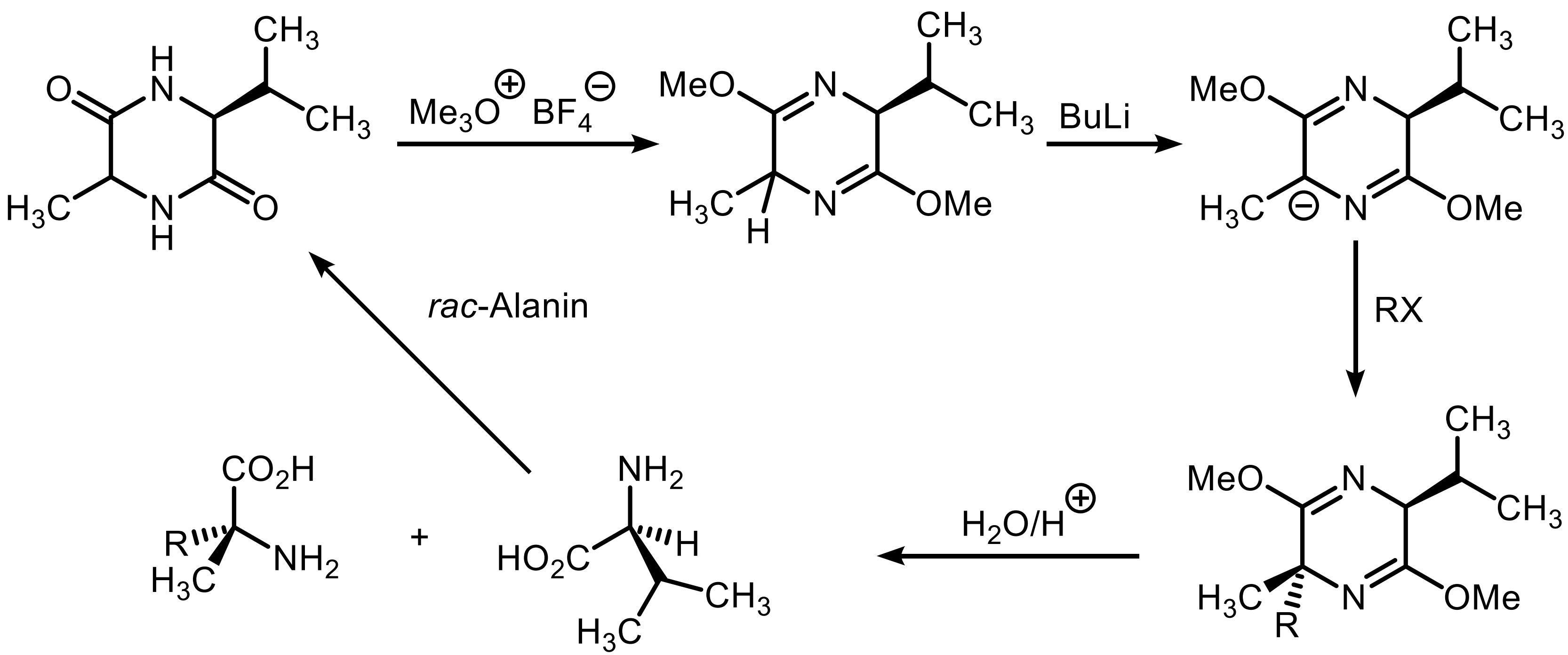
5.2.1.2. Pyrazin

Reaktionen



S_E nur bei aktivierten Pyrazinen

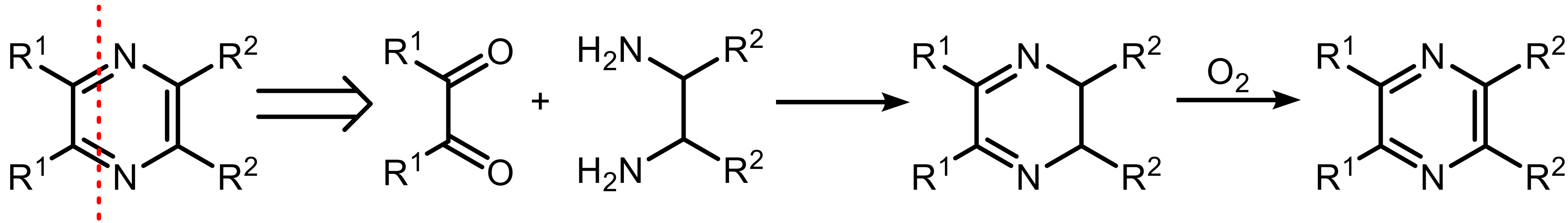
Schöllkopf-Bislactimether-Synthesen

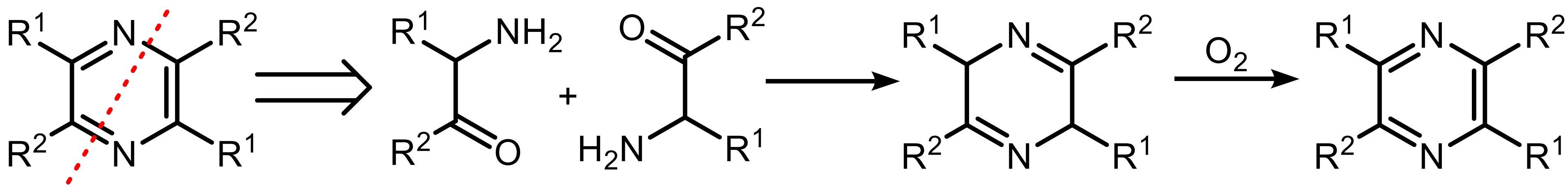


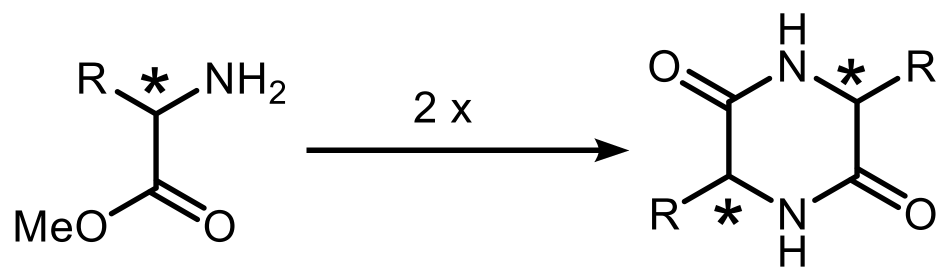
hohe ee-Werte

L-Valin

5.2.1.2. Pyrazin
Synthesen

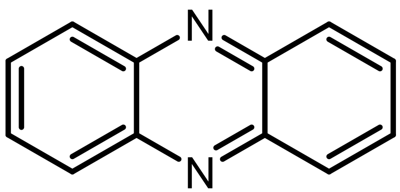


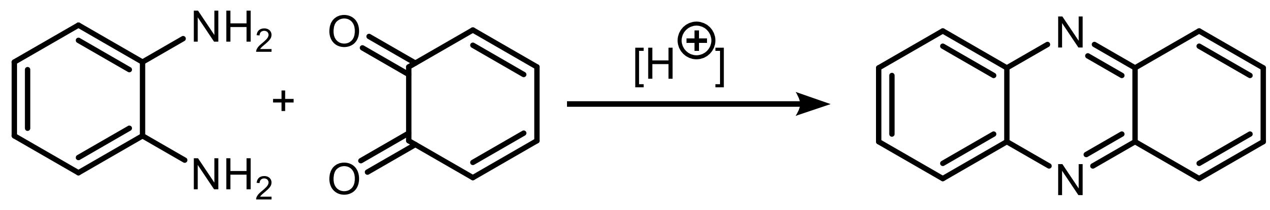




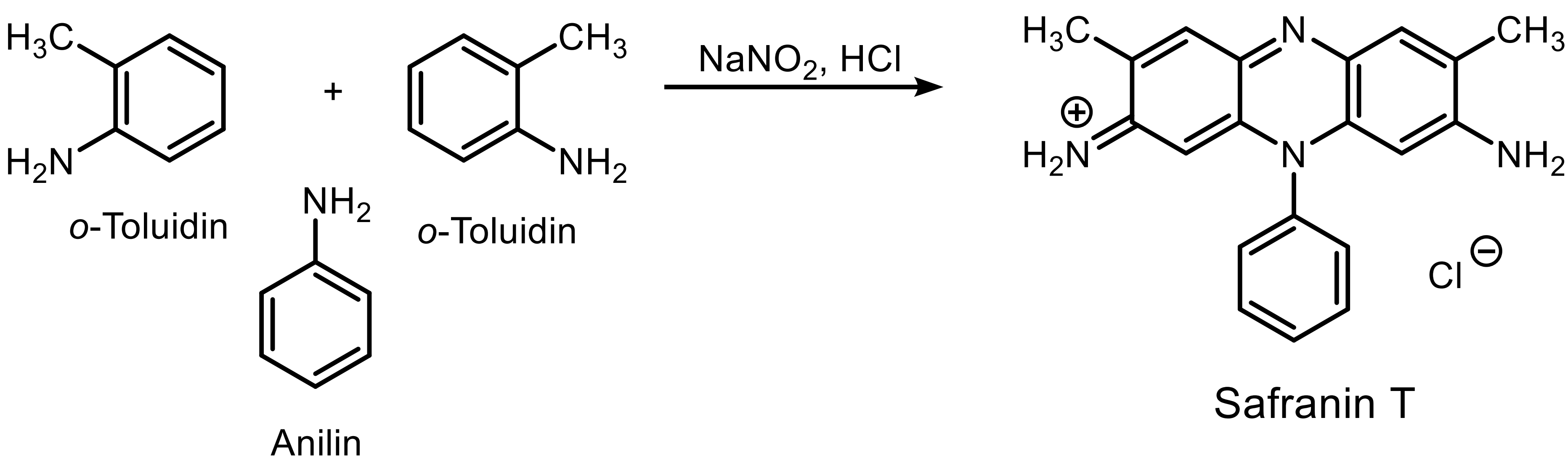
Aminosäure Diketopiperazin

Phenazin

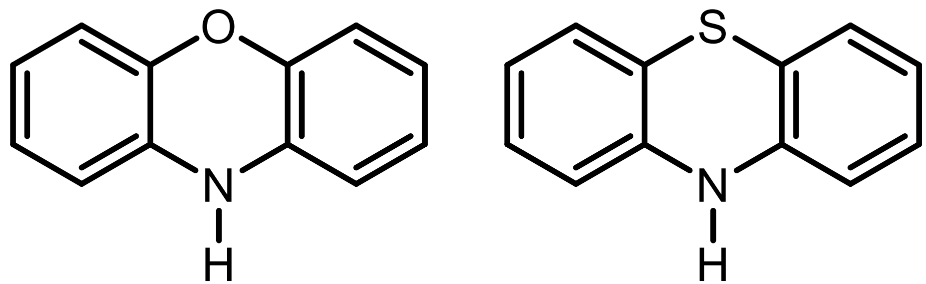




Smp. 171 °C
pK_a (Phenazinium) = 1.23
Grundkörper der Safranine
(Klasse: Beizenfarbstoffe)



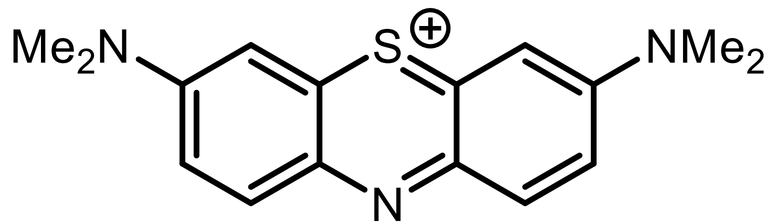
5.2.1.2. Pyrazin
Benzoanellierte Analoga



Phenoxazin Phenothiazin

wegen 16 e⁻ in den Tricyclen (Hückel-Antiaromat)

=> gewinkelt (Butterfly-Struktur); stabil

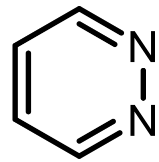


Methylenblau (oxidiertes Phenothiazin)

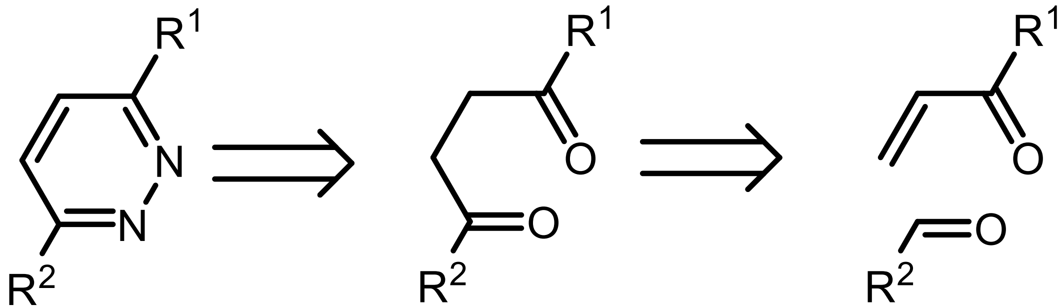
(14 e⁻ im Tricyclus (Hückel-Aromat)

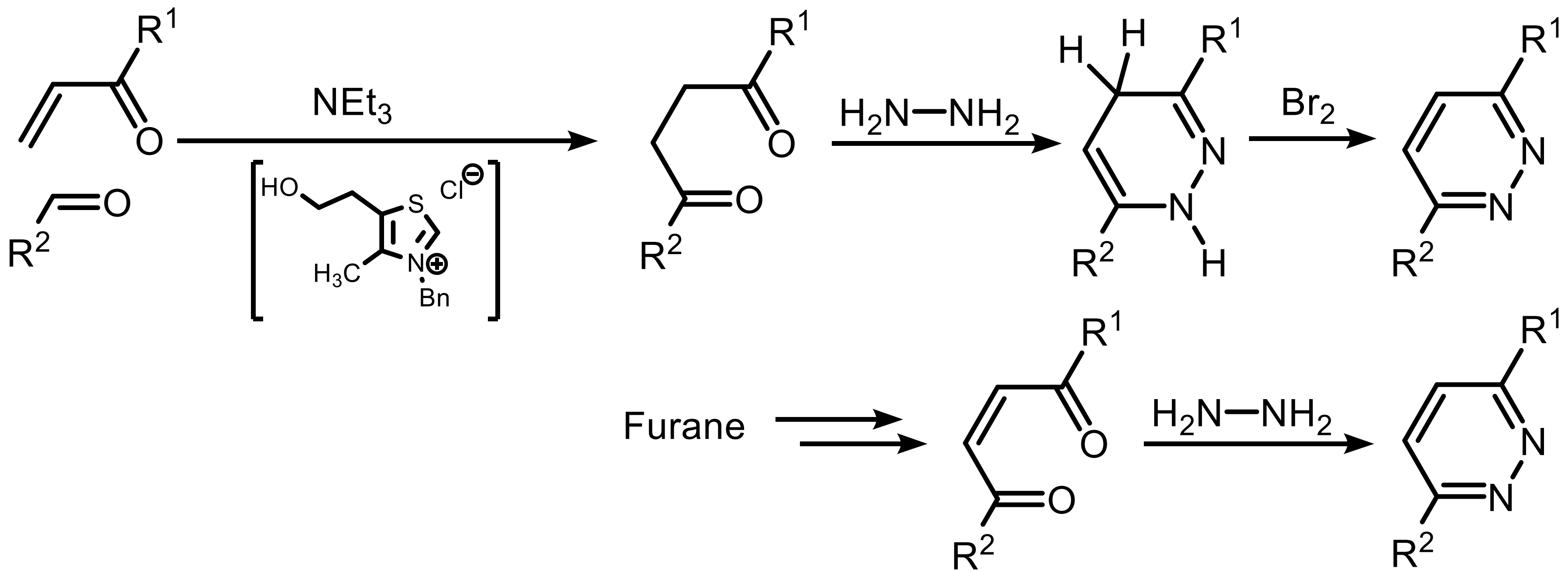
=> planar; stabil)

5.2.1.3. Pyridazin

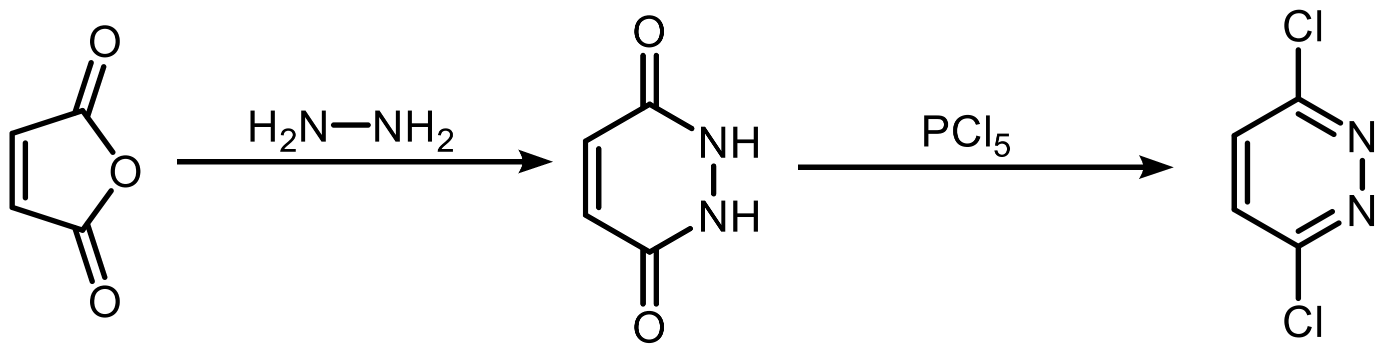


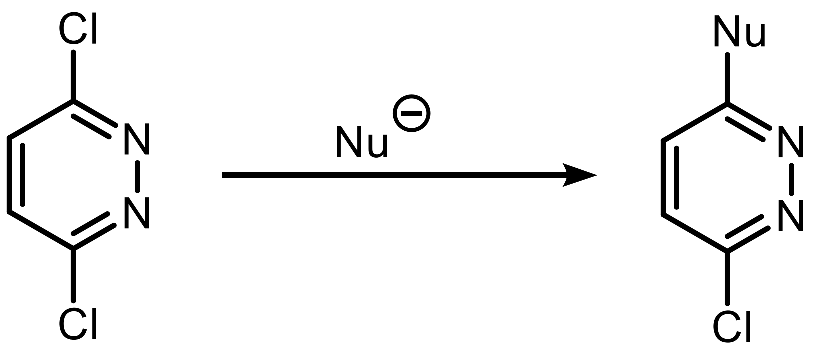
farblose Flüssigkeit
(Sdp. 208 °C)

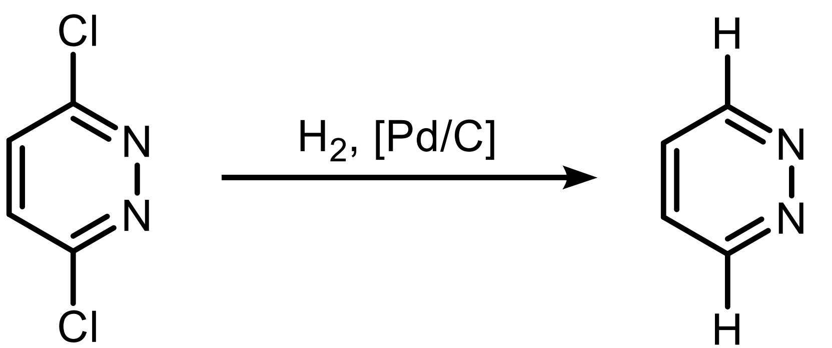




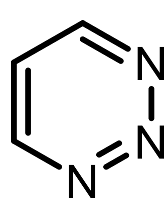
5.2.1.3. Pyridazin



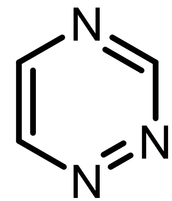




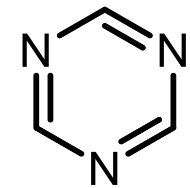
5.3. Sechsring-Heterocyclen mit drei und vier Heteroatomen
5.3.1. Triazin



1,2,3-Triazin

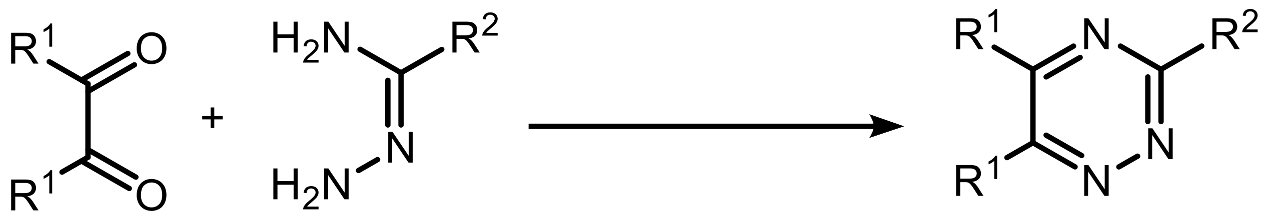


1,2,4-Triazin

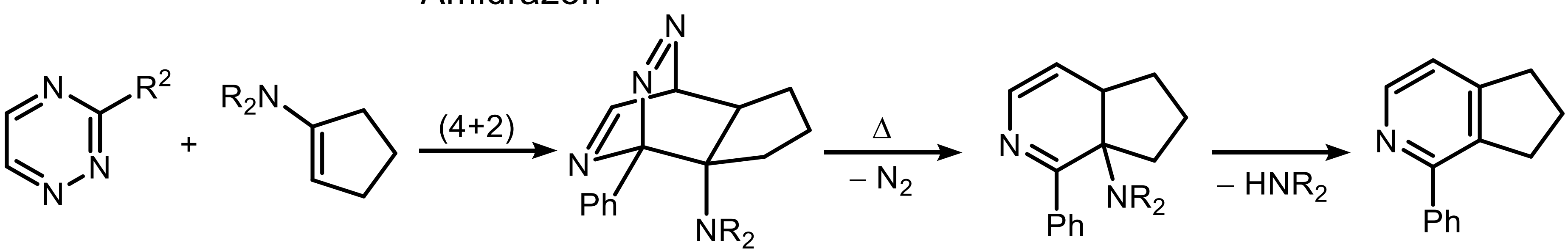


1,3,5-Triazin

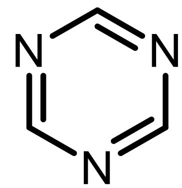
1,2,4-Triazine



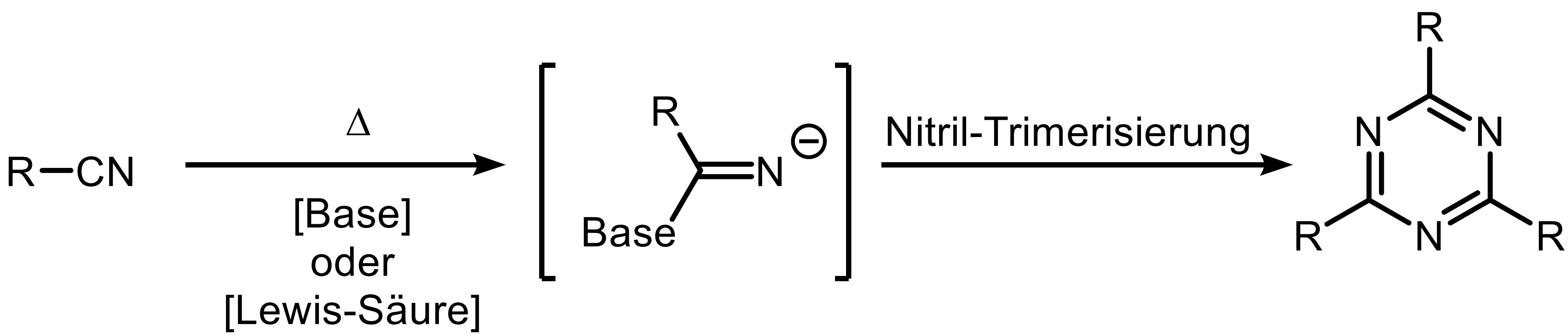
Amidrazon

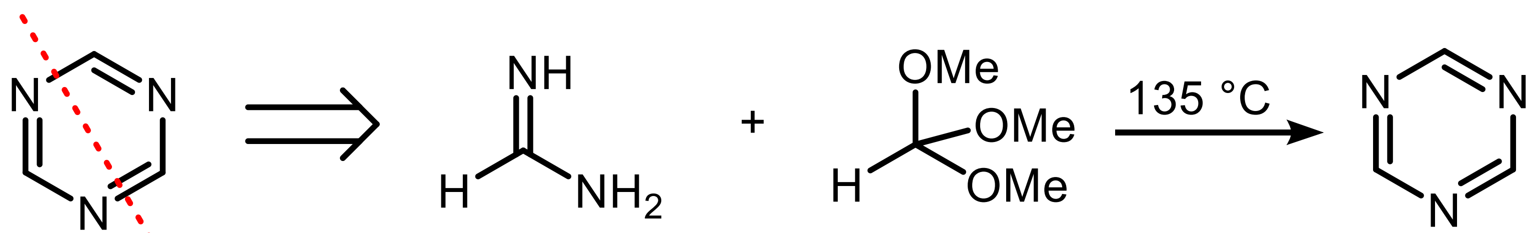


1,3,5-Triazine



Smp. 80 °C
farblos



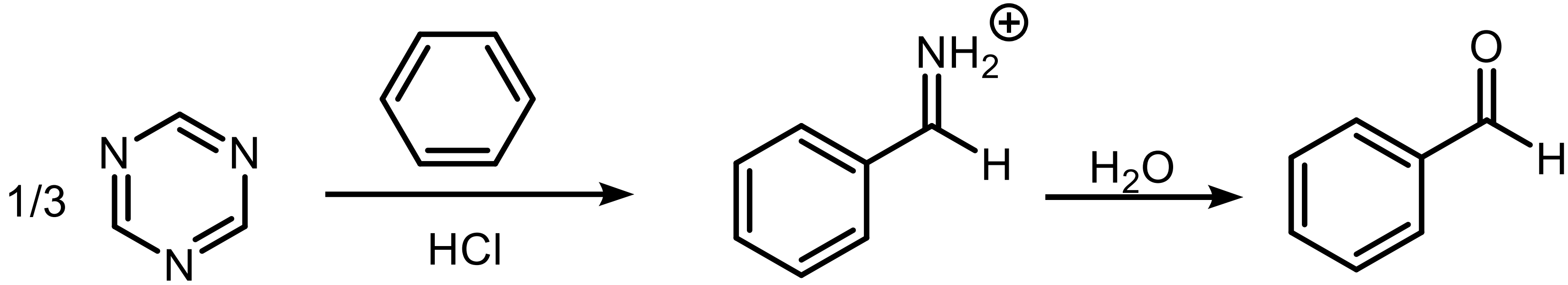


Formamidin

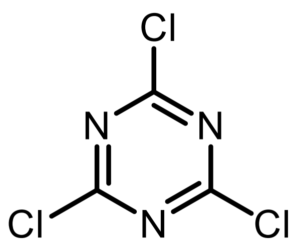
Ameisensäure-
orthoester

Reaktionen

keine Reaktion mit Elektrophilen; mit Nucleophilen erfolgt Ringöffnung (Formylierung)

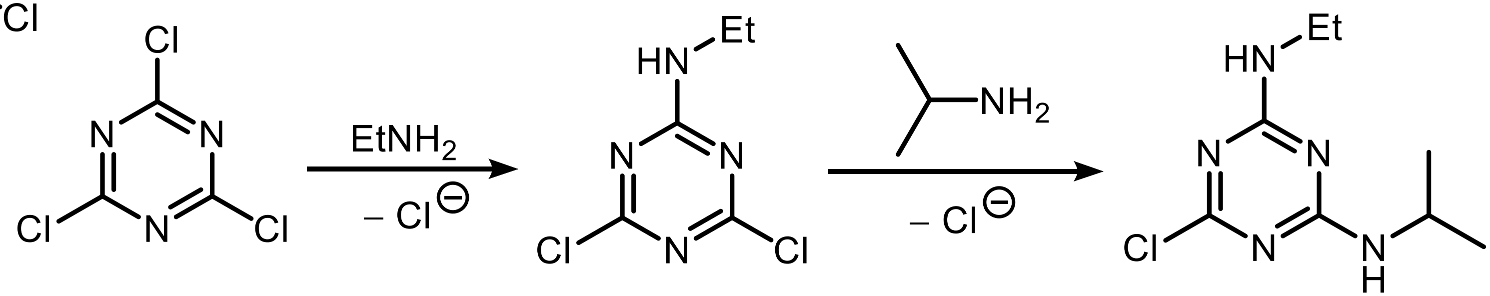


Cyanurchlorid (technische Synthese aus Cl-CN an Aktivkohle)



als Ankergruppe für Farbstoffe, da jede einzelne Cl-Gruppe auch substituiert werden kann;

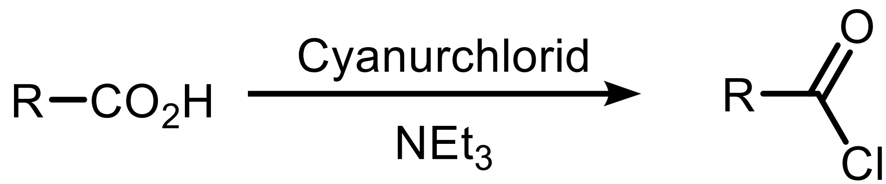
90

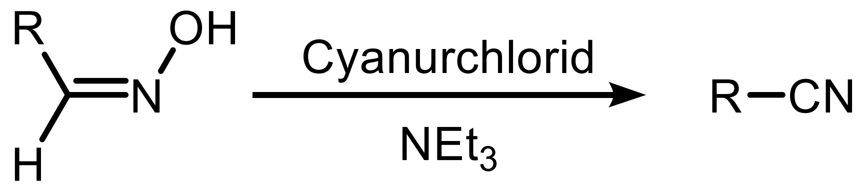


Atrazin
(Herbizid, Photosyntheseinhibitor)

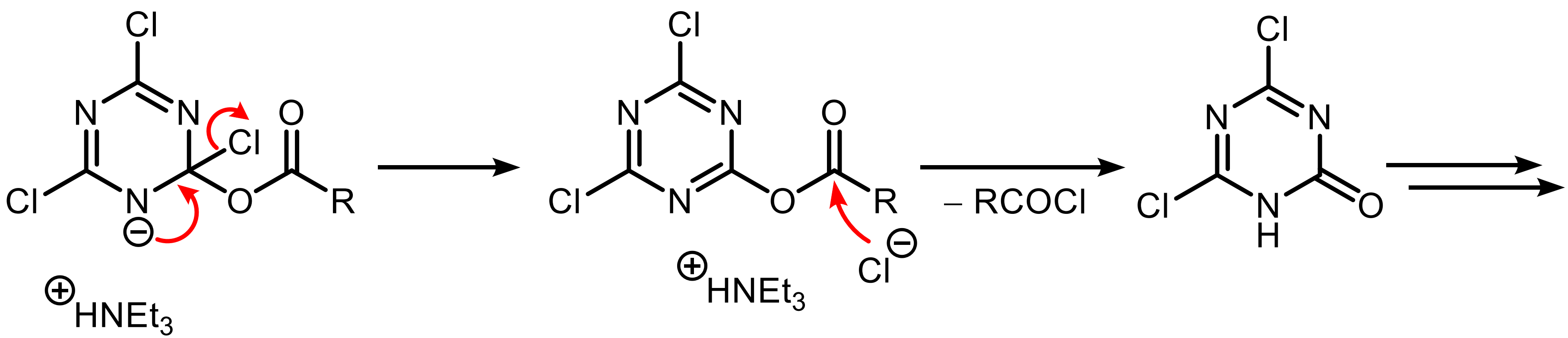
Cyanurchlorid

Darstellung von Säurechloriden/Dehydratisierung

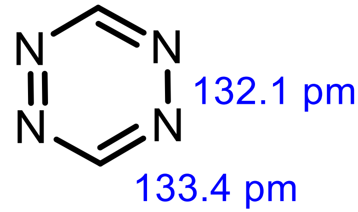




Triebkraft: Bildung von Cyanursäure

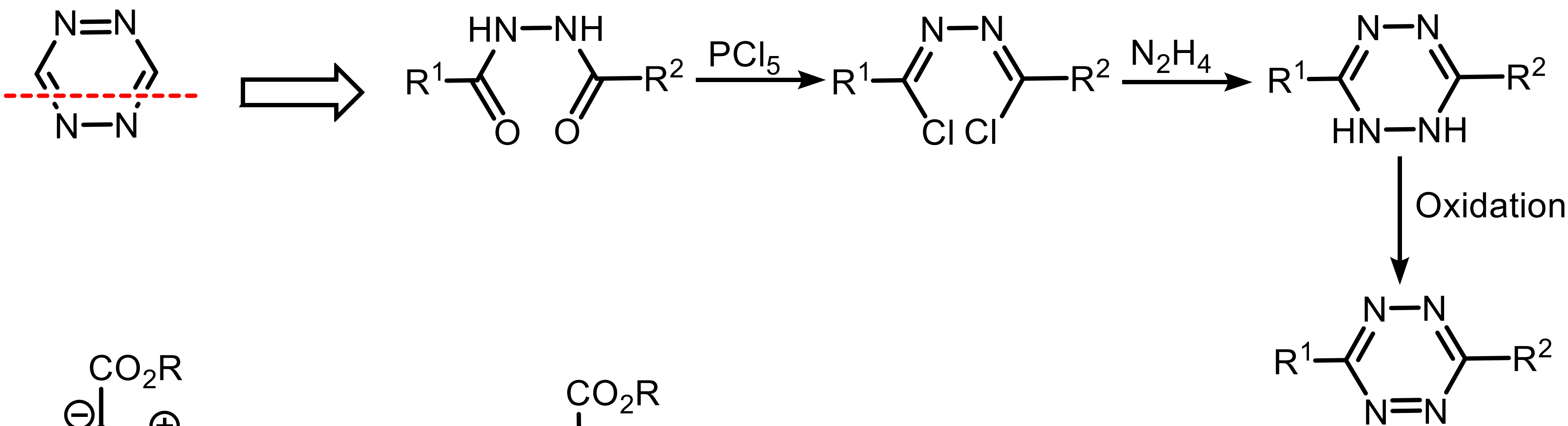


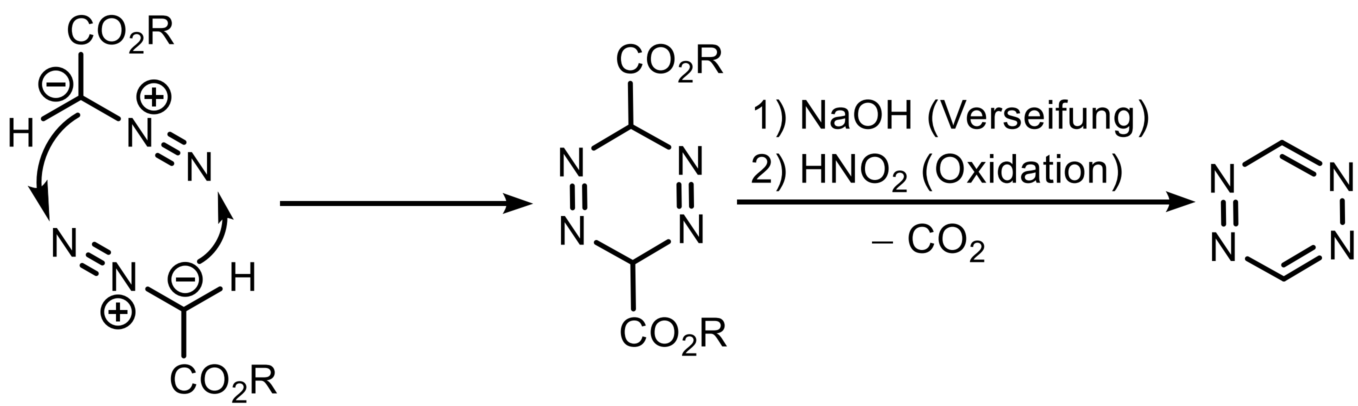
5.3.2. 1,2,4,5-Tetrazin



rotviolett

Synthese: Acylierung von Hydrazin





Reaktionen

Diels-Alder-Reaktionen: Synthese von Pyridazinen